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DUM 2026 CONFERENCE SESSION

SESSION 5: CLIMATE RESILIENCE AND SECURING THE PHYSICAL ASSETS OF DISCOMs

TRANSFORMER RISK INTELLIGENCE

GICs: the unmonitored stressor behind EHV/UHV transformer heating

Presented By

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INTRODUCTION

Heating is the major threat to India's EHV/UHV transformers

1,486 GVA
of 220 kV-and-above transformation capacity (Jul 2026)

28
220 kV+ failures reported to CEA in 2025, only 8 utilities reported

8 of 28
failed within 5 years of a 25–40 year design life

2× ageing
for every +6 °C of hot-spot temperature

TRACKED GLOBALLY

Heatwaves

Record loading on EHV substations: 270.82 GW peak (2026), 53.6 °C (2024)

POORLY INVESTIGATED

Space weather

GIC enters via grounded EHV/UHV neutrals; 234 GIC-conducive events at Jaipur in 2024

LOGGED, NOT SEPARATED

Other origins

Overload, through-faults, insulation and bushing defects, moisture

Our aim: locate heating in EHV/UHV transformers and identify its cause — **heatwave, space weather or another origin** — so utilities take the right precaution at the right substation.

Sources: CEA via Outlook Business (Jul 2026) and Power Line (May 2026); IEC 60076-7; DUM 2026 Session 5 note

From Sun to substation in minutes: how a GIC heats a transformer

1 · Solar eruption

CME / CIR plasma hits Earth's magnetosphere

2 · Geomagnetic storm

Rapid field change (dB/dt) at ground level

3 · Geoelectric field

Drives quasi-DC current along long grounded lines

4 · Neutral entry

GIC enters via grounded EHV/UHV transformer neutrals

5 · Core saturation

Hot-spots, harmonics, VAR demand, relay misoperation

At high latitude it has gone all the way to failure

1989	**Hydro-Québec, Canada** — grid collapsed in 90 seconds; 6 million people lost power
2003	**South Africa** — multiple HV transformers damaged or failed after the Halloween storms
2003	**Malmö, Sweden** — storm-driven outage affected 50,000 customers



Transformer damaged in the March 1989 storm

RELEVANCE

EHV/UHV transformers take the GIC, as heat

Long lines, low-resistance windings and solidly grounded neutrals make 220–765 kV transformers the entry point for GIC. In India they heat rather than fail outright as in polar regions, and that heat stacks on heatwave load.

Heat removes the margin

A transformer at its thermal limit in a heatwave has no headroom. GIC adds hot-spot heating, harmonics and reactive demand on top, and the failure is booked as "overload".

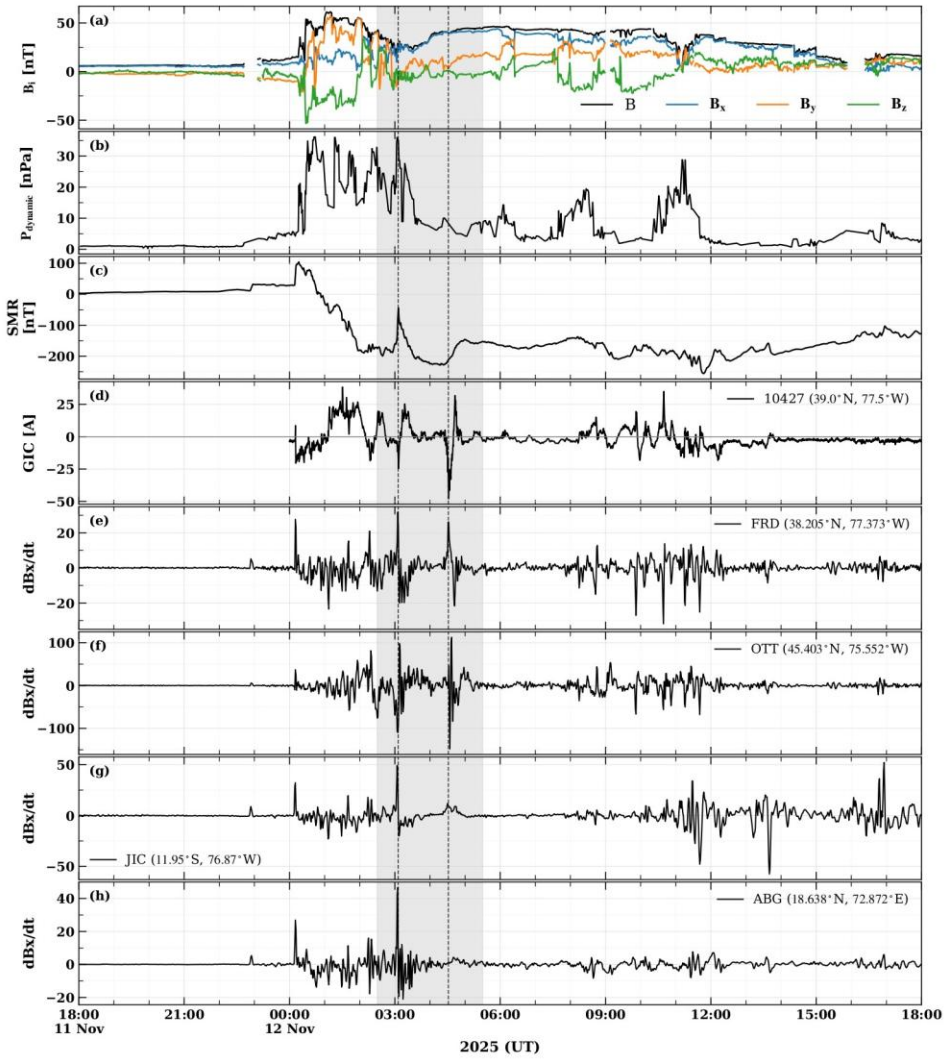
India's exposure is growing

Longer EHV/UHV corridors, a tightly interconnected grid and sharp ground-conductivity contrasts (Deccan basalt, coastal zones) raise induced currents. Risk is set by grid design, not latitude alone.

The electrojet amplifies it

Near the dip equator, the equatorial electrojet and counter-electrojet magnify ground magnetic disturbances (dB/dt), a low-latitude driver that high-latitude GIC models leave out.

USE CASE / CASE STUDY

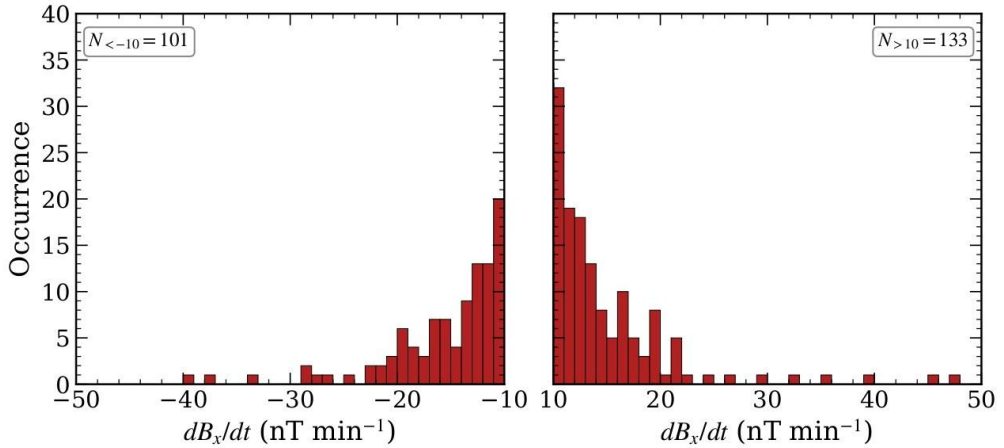


ICME of 12 Nov 2025: solar wind and dB/dt

Case 1 · ICME event, 12 November 2025

- Analysed 1 AU solar-wind data with ground magnetometer records
- Compared U.S. and Indian (Jaipur, Alibag) responses
- Found sharp dB/dt spikes over India, the signature that drives GIC

Case 2 · Jaipur magnetometer, full year 2024



101

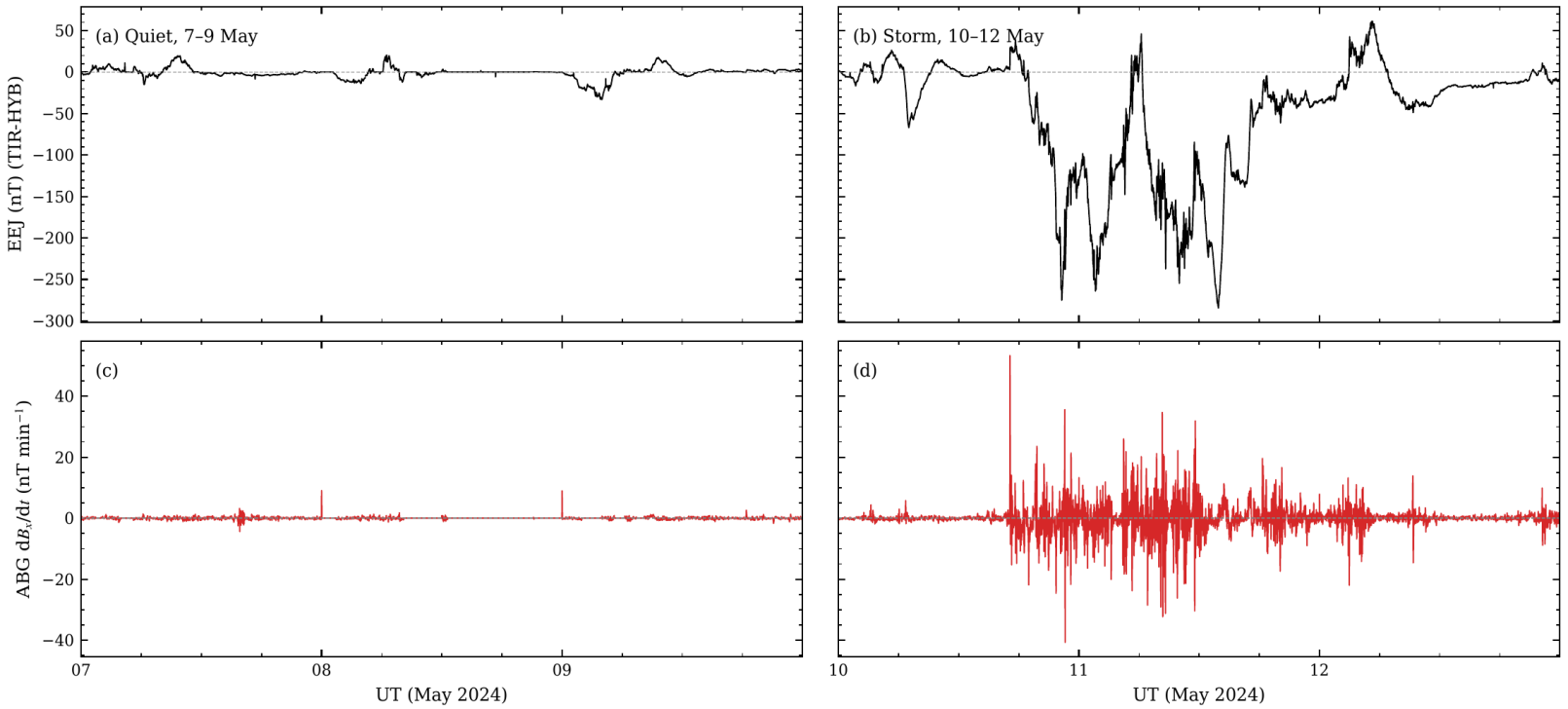
events with dB/dt < −10 nT/min

133

events with dB/dt > +10 nT/min

GIC-driving disturbances do occur over India. What is missing is transformer-side monitoring to measure their impact.

USE CASE / CASE STUDY – MAY 2024 STORM



(a, b) EEJ: Tirunelveli – Hyderabad · (c, d) Alibag dBx/dt · quiet 7–9 vs storm 10–12 May 2024

The storm turned the electrojet into a strong **counter-electrojet** while Alibag dB/dt rose from under 10 to over 50 nT/min.

Quiet days · 7–9 May

EEJ within a few tens of nT; Alibag dB/dt under 10 nT/min

Storm · 10–12 May

≈ –280 nT

EEJ swing: a strong counter-electrojet (CEJ)

≈ 53 nT/min

peak Alibag dB/dt, with storm-level activity for over a day

PRESENTATION ON THE TOPIC

DISTRIBUTION UTILITY MEET DUM 2026

SPACEPROBE

SpaceProbe Transformer Risk Intelligence turns space-weather data and records utilities already hold into risk for each EHV/UHV transformer, **no new hardware, no operational disruption.**

Audit
GIC-proxy, harmonic and thermal anomaly detection

Asset ageing
Stress indices, thermal ageing (FAA), remaining useful life

Reporting
Monthly and quarterly reports, plus post-storm reviews

Grid intelligence
Cross-asset benchmarking and fleet vulnerability ranking

Alert system
Storm watches, CME arrival alerts, real-time risk advisories

Utility data we use, most of it already flows through SCADA

Neutral current	Phase current	Voltage	Active power	Reactive power
Power factor	Oil / winding / hot-spot temp.	OLTC tap position	Relay logs	DGA history

Combined with: solar-wind data at L1, CME/ICME tracking, ground magnetometer dB/dt and geomagnetic indices, read by a team with 17+ years of research on the drivers of GIC.

KEY TAKEAWAYS / RECOMMENDATIONS

1	**Nobody is measuring GIC in India**	No nationwide operational GIC monitoring; Jaipur alone logged 234 GIC-conducive events (|dB/dt| > 10 nT/min) in 2024.
2	**Treat GIC as a root-cause candidate**	Overlay storm days (Kp, dB/dt) on 220 kV+ transformer heating, tripping and failure records before closing them as "overload".
3	**Use the already collected data**	Neutral current, phase current, reactive power, load voltage, harmonics, temperature and DGA give a GIC-proxy screen without new hardware.
4	**Add a geomagnetic layer to risk maps**	Rank EHV/UHV substations by line length, grounding and ground conductivity next to IMD heat, cyclone and flood hazards.
5	**Put space weather in preparedness plans**	Link CME arrival alerts to Heat Action Plans: pre-emptive load transfer, derating and spares when a storm meets peak heat.
6	**Start small, measure, then scale**	Pilot neutral-current monitoring at high-exposure sites and let measured data decide the wider rollout.
7	**Pilots separate the sources**	At EHV/UHV substations, correlating temperature, neutral current and harmonics with heatwave and space-weather data quantifies each contributor.

We are currently looking for: A 3–5 EHV/UHV substation pilot with SpaceProbe, screening on existing data, then neutral-current monitoring.

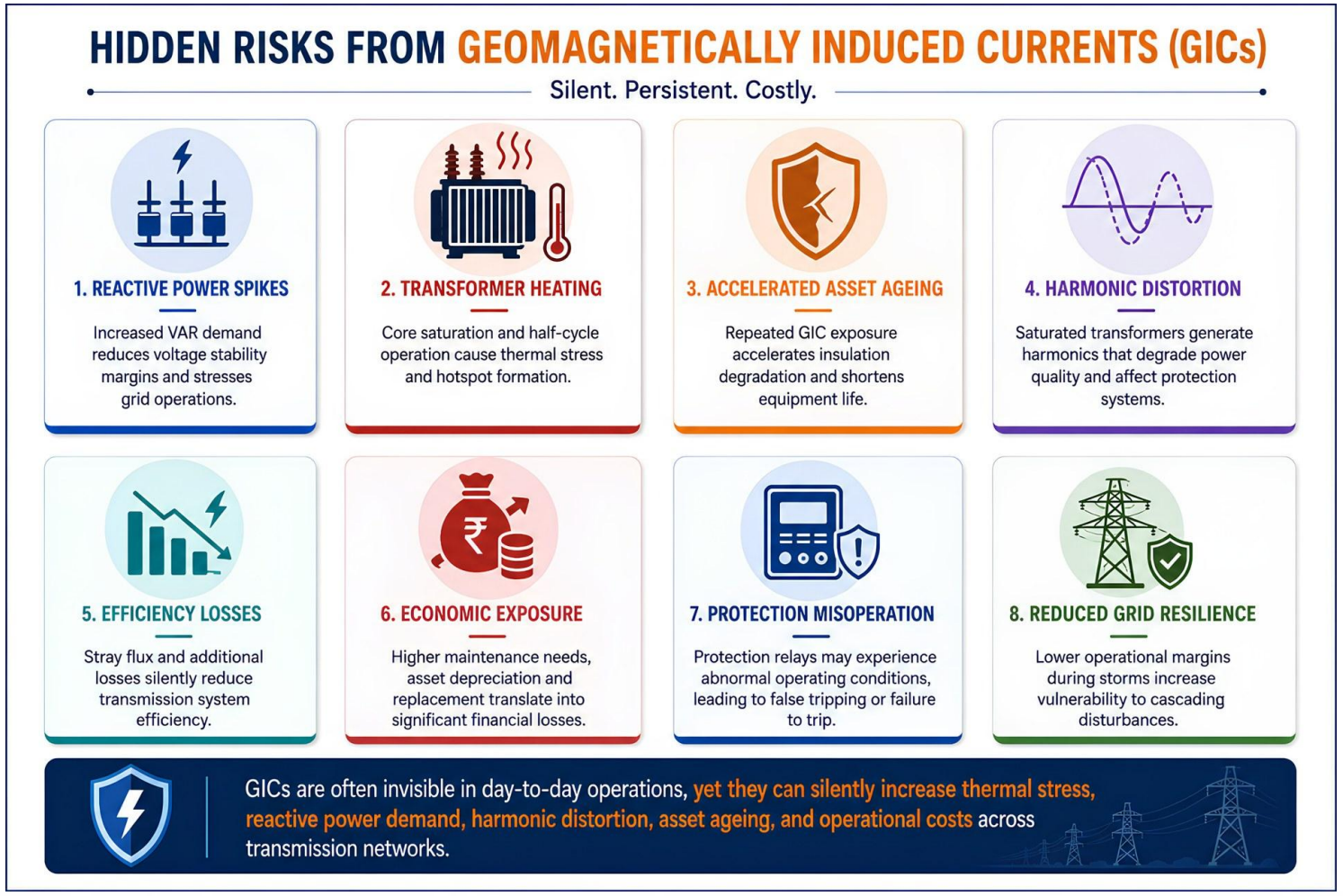
THANK YOU

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SPACEPROBE





Who monitors GIC today?

United States — operational GIC and neutral-current monitoring

Canada — utility-wide monitoring since the 1989 blackout

Nordic countries & UK — national monitoring networks

New Zealand — monitoring of vulnerable transmission assets

China — growing GIC monitoring on UHV networks

> **India:** no nationwide operational GIC monitoring yet.